

TimescaleDB Time-Series Operations & Telemetry Guide

Platform: CloudPulse Autonomous FinOps | Engine: PostgreSQL 18.6 + TimescaleDB 2.23.0 | Date: September 2026

Executive Summary: CloudPulse utilizes TimescaleDB hypertables to ingest high-frequency, sub-second telemetry streamed from guest virtual machines alongside CloudWatch hypervisor metrics. This runbook provides the operational commands to inspect the database, verify hypertable partitioning, execute sub-millisecond time-bucket rollups, and validate live ingestion from running EC2 instances.

1. Option 1: 1-Line Terminal CLI Command (Automated Inspection)

Run the automated diagnostic CLI tool directly from the repository root to verify TimescaleDB connectivity, hypertables, active chunks, and sample time-bucket rollups in a single step:

```
export DATABASE_URL="postgresql://cloudpulse_db_i402_user:LkQtWiawBtDkRyCNlaQ06C1agZi6XZ1d@\
dpg-dam114e1egvs73c1lhpg-a.oregon-postgres.render.com/cloudpulse_db_i402"

./venv/bin/python scripts/show_timeseries_db.py
```

Standard Output Verification:

- **TimescaleDB Extension:** Reports active version 2.23.0 on PostgreSQL 18.6.
- **Hypertables & Chunks:** Confirms resource_telemetry (2 chunks) and daily_spend_records (1 chunk).
- **Real In-Guest Rows:** Live CPU (0.0% – 51.2%), RAM (12.6%), and Disk (30.5%) records from i-07d01b00f95a4cc41.
- **Time-Bucket Rollup:** Computes 1-minute analytical aggregations (avg, peak, sample counts).

2. Option 2: Connect via psql (Interactive PostgreSQL Shell)

For direct database administration, connect via the official PostgreSQL client:

```
psql "postgresql://cloudpulse_db_i402_user:LkQtWiawBtDkRyCNlaQ06C1agZi6XZ1d@\
dpg-dam114e1egvs73c1lhpg-a.oregon-postgres.render.com/cloudpulse_db_i402"
```

Diagnostic Query 1: Verify TimescaleDB Extension

```
SELECT extname, extversion FROM pg_extension WHERE extname = 'timescaledb';
```

Diagnostic Query 2: List All Active Hypertables

```
SELECT hypertable_schema, hypertable_name, num_dimensions, num_chunks, compression_enabled
FROM timescaledb_information.hypertables;
```

Diagnostic Query 3: Inspect Hypertable Chunks & Partition Time Ranges

```
SELECT chunk_name, hypertable_name, range_start, range_end
FROM timescaledb_information.chunks;
```

Diagnostic Query 4: View Latest High-Frequency Raw Telemetry Ticks

```
SELECT time, resource_id, metric_name, val_avg, val_max, val_p95
FROM resource_telemetry
ORDER BY time DESC
LIMIT 10;
```

Diagnostic Query 5: Real-Time 1-Minute Time-Bucket Analytical Rollup

Aggregates second-by-second guest agent telemetry into continuous 1-minute time windows:

```
SELECT
    time_bucket('1 minute', time) AS bucket,
    metric_name,
    ROUND(AVG(val_avg)::numeric, 2) AS avg_value,
    ROUND(MAX(val_max)::numeric, 2) AS peak_value,
    COUNT(*) AS sample_count
FROM resource_telemetry
WHERE resource_id = 'i-07d01b00f95a4cc41'
GROUP BY bucket, metric_name
ORDER BY bucket DESC
LIMIT 12;
```

3. Option 3: REST API Integration Endpoints (cURL / HTTP)

TimescaleDB metrics are exposed through the CloudPulse REST API gateway:

```
# 1. Check TimescaleDB connection health and engine version
curl -s https://cloud-cost-optimization.onrender.com/api/v2/database/status

# 2. Query high-frequency hypertable records directly
curl -s "https://cloud-cost-optimization.onrender.com/api/v2/telemetry/hypertable?limit=10"
```

4. Key TimescaleDB Concepts in CloudPulse

TimescaleDB extends PostgreSQL with dedicated primitives engineered for multi-tenant cloud time-series telemetry:

Concept	Implementation in CloudPulse	Operational & Architectural Benefit
Hypertable	resource_telemetry (7-day interval) daily_spend_records (30-day interval)	Presents a unified standard PostgreSQL table interface while auto-partitioning writes into physical chunk tables.
Chunks	Partitioned by time intervals (e.g., 7 days per telemetry chunk)	Keeps indexes in memory, preventing B-tree index bloat and guaranteeing constant-time write throughput at scale.
time_bucket()	High-speed rollup primitive for 1m, 5m, 1h, and 24h charts	Replaces slow date_trunc with optimized C-level arithmetic, enabling sub-millisecond multi-day aggregations.

Verified Production Database Topology:

- **Instance ID:** dpg-dam114e1egvs73c1lhpg-a | **Name:** cloudpulse-db (Oregon)
- **Database Name:** cloudpulse_db_i402 | **Postgres Engine:** Version 18.6 | **Timescale:** v2.23.0
- **Status:** Healthy, live connections accepted from Render API & authenticated CLI.